

Metro Ticket Generating System in ServiceNow

Category: Transportation · Individual Project

1. Description

Digitizes metro ticket booking end-to-end using ServiceNow — station/date selection, automated fare calculation, instant QR ticket generation, and email confirmation, with dashboards for operations visibility.

2. Problem Statement

Manual, counter-based ticketing causes long queues, inconsistent fare pricing, and zero visibility into demand or revenue for operations teams.

3. Objectives

- Remove manual ticketing entirely
- Guarantee consistent, automated fare pricing
- Issue an instant digital (QR) ticket
- Give operations real-time booking/revenue visibility

4. Methodology — 5 Phases

Phase	Focus	Deliverable
1	Requirement Analysis & Planning	Scope + roadmap
2	Backend Development & Configurations	Data model + automation
3	UI/UX Development & Customization	Booking portal
4	Data Migration, Testing & Security	Tested, secured system
5	Deployment, Documentation & Final Presentation	Live system + docs + demo

5. Technology Stack

Layer	Technology
Frontend	Service Portal, Service Catalog, Client Scripts, UI Policies
Backend	Business Rules, Script Includes, Flow Designer
Database	Custom table <code>u_metro_ticket</code>
Integration	External QR generation API
Reporting	Reports & Performance Analytics
Notifications	Notification Engine (email)

6. Assumptions & Constraints

- Single ServiceNow developer instance, no external payment gateway
- Fixed (not dynamic) fare mapping between stations
- Online-only QR validation (no offline mode)

7. Roles Involved

- **Passenger** — books tickets
- **Operations Team** — monitors dashboards
- **ServiceNow Admin** — configures & maintains platform
- **Support Staff** — manages/resolves ticket issues

8. Skills Required

- ServiceNow platform administration
- ServiceNow CSA / CAD certifications
- Service Portal / UI component development

9. Project Links

-  Demo Link: *add link here*
-  GitHub Link: *add link here*

10. Project Stats

Metric	Value
Total Epics	6
Total Tasks	26
Total Subtasks	12
Overall Progress	0%

Notion Structure (Sub-pages)

Phase 1: Requirement Analysis & Planning

Phase 2: Backend Development & Configurations

Phase 3: UI/UX Development & Customization

Phase 4: Data Migration, Testing & Security

Phase 5: Deployment, Documentation & Final Presentation

Project Conclusion